

DIGITAL COLONIALISM IN THE ASAL CORRIDOR

A Feminist Analysis of Technology Challenges Facing Indigenous and Pastoralist Communities in Northern Kenya

Prepared by: Beyond Borders Indigenous Feminist Tech Collective

Region: Turkana, Samburu, Marsabit, Isiolo, Wajir, and Mandera Counties, Northern Kenya

Period: Research conducted 2024 - 2025

Methodology: Community consultations (450+ participants), participatory mapping, digital security audits, infrastructure surveys

Communities: Turkana, Samburu, Borana, Rendille, Gabra, Somali pastoralist communities

NOTE: This report is not a plea for connectivity. It is a feminist interrogation of *who controls technology, who profits from it, and whose bodies are made visible or invisible by its design*. We write from the position that indigenous pastoralist communities in Northern Kenya are not technologically backward—they are technologically *dispossessed*.

EXECUTIVE SUMMARY

Northern Kenya's ASAL corridor is one of the most technologically marginalized regions in East Africa. Yet this marginalization is not accidental. It is the result of deliberate infrastructure neglect, extractive digital colonialism, and patriarchal control over the few technologies that do reach nomadic communities. This report documents the specific, intersectional tech challenges facing indigenous pastoralist communities—and particularly indigenous young women—in the Northern Kenya corridor.

Our analysis moves beyond the simplistic narrative of "digital divide" to examine how technology in the ASAL is actively used to surveil, extract, and exclude. We document six interconnected challenge domains: infrastructure apartheid, extractive data colonialism, patriarchal tech control, climate-tech failure, educational exclusion, and the violence of humanitarian innovation. For each, we offer community-generated responses and a framework for indigenous feminist tech sovereignty.

6 Challenge domains identified	450+ Community consultations	0% Community consent for data extraction by state/NGOs
92% Young women with no private device access	78% Settlements with no grid electricity	100% Reports of external tech designed without community input

1. INFRASTRUCTURE APARTHEID: THE PHYSICS OF EXCLUSION

1.1 Connectivity as a Colonial Geography

The ASAL corridor covers 23 counties and over 80% of Kenya's land area, yet less than 12% of the national fiber optic network reaches these regions. Mobile network coverage is concentrated along tarmac roads and urban centers; vast pastoralist rangelands have no signal at all. In our surveys, 78% of nomadic settlements reported having no reliable mobile data, and 64% had no voice signal during the dry season when communities move to remote grazing areas.

This is not a technical failure. It is an **economic decision** by telecom companies and the Kenyan state. Pastoralist populations are deemed unprofitable by market logic. The Universal Service Fund, which is supposed to subsidize rural connectivity, has been plagued by corruption and mismanagement, with less than 3% of collected funds directed to ASAL infrastructure. The result is that indigenous communities are digitally invisible—not because they reject technology, but because the state and private sector have decided their connectivity is not worth the investment.

1.2 The Electricity Gap

Electricity access in ASAL counties averages 8–15%, compared to 65% nationally. For nomadic communities, this figure drops to near zero. Families rely on car batteries, small solar lanterns, or long walks to town centers for phone charging. A single smartphone charge can cost 50 Kenyan shillings—equivalent to a day's worth of water. This means that technology use is rationed: phones are turned off for days, content cannot be downloaded, and digital participation is limited to emergency use only.

For young women, the electricity gap has specific gendered consequences. Charging a phone requires access to a charging station, which is often in a town center where young women face harassment, or in a male-controlled household where phone use is monitored. We documented cases where young women were denied access to the family solar panel because their fathers considered phone charging a waste of resources for girls.

1.3 Device Apartheid

The devices available in ASAL markets are the cheapest, lowest-quality smartphones imported from China and India—often refurbished units with failing batteries, minimal storage, and outdated Android versions that cannot run modern apps. These devices are not designed for dusty, hot, rough environments. Average device lifespan in our communities is 8–14 months, after which repair is impossible due to lack of spare parts and technical skills.

More critically, these devices are **shared**. In our consultations, 92% of young women reported having no private device. Phones belong to fathers, husbands, or brothers. This means that every digital action is surveilled. A saved PDF, a WhatsApp message, or a browser history is visible to the entire household. For young women seeking information about FGM, contraception, or escape from

child marriage, this shared device structure is not an inconvenience—it is a structural barrier to bodily autonomy.

2. EXTRACTIVE DATA COLONIALISM: HARVESTING INDIGENOUS BODIES

2.1 Biometric Harvesting by Humanitarian Agencies

Northern Kenya hosts some of the world's most intensive humanitarian biometric data collection. Refugees, IDPs, and drought-affected pastoralists are required to provide fingerprints, iris scans, and facial recognition data to access food aid, medical services, and education support. This data is collected by INGOs, UN agencies, and government departments using systems built by Global North tech companies.

In our consultations, **not a single community member** reported being asked for meaningful consent. They were told: no biometric data, no food. The power asymmetry is absolute. Communities do not know where their data is stored, who has access, or how long it is retained. When we asked UNHCR and WFP staff about data deletion protocols, they could not provide clear answers. Indigenous pastoralists have become some of the most biometrically surveilled populations on earth, with none of the legal protections afforded to citizens in the Global North.

2.2 The Research Extraction Economy

Every year, dozens of researchers, PhD candidates, and NGO consultants descend on Northern Kenya with cameras, drones, and survey apps. They photograph indigenous women, record their stories, and upload the data to universities in Europe and North America. This research is rarely shared back with communities. It is published in paywalled journals, presented at conferences, and used to secure grants that never reach the people photographed.

We call this the **research extraction economy**. It operates on the same logic as colonial resource extraction: indigenous knowledge is treated as raw material to be harvested, processed elsewhere, and sold for profit. When we surveyed community members about whether they had ever seen a research report about their own lives, 97% said no. When we asked if researchers had ever returned to share findings, 99% said no. This is not research partnership. It is digital colonialism.

2.3 Platform Colonialism

The few digital platforms that do reach ASAL communities are foreign-owned and extractive. Facebook and WhatsApp are the primary communication tools, but they harvest metadata, sell attention to advertisers, and moderate content using algorithms that do not understand local languages or cultural contexts. In 2024, Facebook's content moderation system removed posts by Samburu women discussing FGM survivor support, classifying them as harmful content—while allowing misogynistic hate speech in Swahili to remain. The platforms control the infrastructure, the rules, and the profits. Communities control nothing.

3. PATRIARCHAL TECH CONTROL: GENDER, DEVICES, AND DOMINATION

3.1 The Male Gatekeeper Economy

In Northern Kenya's patriarchal pastoralist societies, technology is not gender-neutral. It is actively controlled by men as a domain of power. Phone ownership is overwhelmingly male. Internet cafes are male spaces. Tech repair shops are run by men. Community radio stations are managed by male elders. The digital public sphere is, in practice, a male sphere.

Young women who seek digital access must navigate a gauntlet of male gatekeepers: fathers who control phone charging, brothers who monitor WhatsApp, shopkeepers who refuse to serve unaccompanied women, and community leaders who frame female phone use as moral corruption. In our consultations, 68% of young women reported having a phone confiscated by a male relative. 45% reported being beaten or threatened for accessing information about contraception or legal rights online.

3.2 Tech-Facilitated Gender-Based Violence

The same technologies that exclude women are also used to harm them. We documented extensive tech-facilitated GBV in ASAL communities: image-based abuse where intimate photos are shared in WhatsApp groups as punishment for disobedience; SIM swap attacks where male relatives take over women's phone numbers to isolate them from support networks; and organized harassment campaigns on Facebook targeting women who speak out against FGM or child marriage.

The response from platforms and authorities is absent. Facebook's reporting mechanisms require English literacy and stable internet—both unavailable to most survivors. Kenyan police in ASAL regions dismiss digital abuse as a family matter. There are no shelters with digital security support, no paralegals who understand SIM swaps, and no community-based tech support for survivors. The violence is enabled by the same infrastructure gap that excludes women from digital power.

3.3 The Erasure of Indigenous Women's Tech Knowledge

Perhaps the deepest patriarchal tech challenge is the erasure of indigenous women's existing technological knowledge. Pastoralist women have managed complex ecological information systems for generations—tracking water sources, pasture quality, animal health, and climate patterns across vast territories. This is sophisticated data work. Yet it is never recognized as technology. When external tech actors arrive with GPS apps and satellite data, they treat elder women as ignorant beneficiaries, not as expert knowledge holders. This erasure is both patriarchal and colonial: it devalues women's knowledge because it is embodied, oral, and indigenous, rather than coded, written, and imported.

4. CLIMATE-TECH FAILURE: WHEN INNOVATION MEETS DISPOSSESSION

4.1 Inappropriate Tech for Climate Realities

Climate adaptation tech deployed in Northern Kenya is overwhelmingly designed by urban engineers with no understanding of pastoralism. Satellite-based early warning systems send SMS alerts about drought—but to phones that are often off, shared, or out of signal range. Mobile money platforms for drought insurance require KYC documentation that nomadic communities do not possess. Solar pumps for irrigation assume stationary farming, not mobile pastoralism.

We documented a case in Turkana where a NGO installed a solar-powered water monitoring system at a cost of 2.3 million Kenyan shillings. Within six months, the panels were covered in dust, the sensors had been damaged by livestock, and the community had no training to maintain the system. The NGO staff had left. The equipment now sits unused. This is not an isolated incident. It is the standard model of climate-tech intervention in the ASAL: expensive, inappropriate, externally controlled, and abandoned.

4.2 The Commodification of Indigenous Ecological Knowledge

Climate-tech actors are increasingly interested in indigenous ecological knowledge—but not in partnership. Tech startups and research institutions want to digitize pastoralist knowledge of pasture, water, and weather patterns to feed into global climate models and carbon credit schemes. They want the data, but not the data holders. There is no framework for indigenous communities to retain intellectual property rights over their ecological knowledge once it is digitized. There is no mechanism for benefit-sharing. The knowledge that has sustained pastoralism for millennia is being extracted for global climate markets, with zero return to the communities who created it.

5. EDUCATIONAL EXCLUSION: THE DIGITAL SKILLS GAP AS STRUCTURAL VIOLENCE

5.1 Schools Without Tech, Tech Without Schools

ASAL regions have the lowest school enrollment rates in Kenya, with girls' enrollment dropping to under 30% in some pastoralist areas. Where schools exist, they lack electricity, computers, and trained teachers. The government's Digital Literacy Program, which promised tablets for all primary students, has barely reached ASAL schools. Where tablets were delivered, they arrived without charging infrastructure, without relevant content, and without teachers who knew how to use them.

The result is a generation of indigenous young women who are completely excluded from digital skills training. They cannot code. They cannot manage servers. They cannot design apps. Not because they lack capacity, but because the education system has systematically denied them access. When tech opportunities do arise—coding bootcamps, innovation hubs, startup competitions—they are located in Nairobi, require English fluency, and assume a level of formal education that pastoralist girls do not have. The digital skills gap is not a personal failing. It is structural exclusion.

5.2 The Language of Tech

All major tech platforms, coding languages, and digital tools are in English. While Swahili is widely spoken, the specific languages of Northern Kenya—Turkana, Samburu, Borana, Rendille, Gabra—have zero presence in digital infrastructure. There are no keyboards in these languages. No voice assistants. No auto-correct. No content. For young women who speak these languages as their mother tongue, the digital world is not just inaccessible—it is linguistically alienating. It tells them: your language does not belong in the future.

6. THE VIOLENCE OF HUMANITARIAN INNOVATION

6.1 The Innovation Industrial Complex

Northern Kenya has become a testing ground for humanitarian technology. Drones, blockchain, AI prediction models, and biometric ID systems are piloted on pastoralist populations with minimal consent and no accountability. The logic is simple: ASAL communities are poor, dependent on aid, and lack the political power to refuse. They are the perfect subjects for tech experimentation.

We call this the **innovation industrial complex**. It operates through a cycle of extraction: a tech company or NGO identifies a "problem" in Northern Kenya; designs a solution without community input; pilots it for 12–18 months; collects data and success stories; wins awards at international conferences; and then abandons the project when funding ends. The community is left with broken equipment, lost trust, and no capacity to maintain or adapt what was built. The innovator moves on to the next "pilot."

6.2 The Myth of Participation

Every humanitarian tech project claims to be participatory. But our research reveals that participation in the ASAL is typically performative: a few community members are invited to a workshop in a town center, given transport reimbursement, and asked to validate pre-designed solutions. There is no meaningful community control over design, data, or governance. There is no transfer of technical skills. There is no ownership. When we asked community members whether they felt they had genuine decision-making power in tech projects affecting their lives, 94% said no.

7. COMMUNITY-GENERATED RESPONSES: TOWARD INDIGENOUS FEMINIST TECH SOVEREIGNTY

Despite these overwhelming challenges, indigenous pastoralist communities—particularly young women—are not passive victims. They are actively developing strategies of resistance, adaptation, and autonomous tech creation. This section documents community-generated responses that point toward a model of indigenous feminist tech sovereignty.

- **Offline-First Networks:** Communities are creating mesh networks, Bluetooth sharing circles, and SD-card-based content libraries that function entirely without internet. These networks are controlled by community members, not telecom companies.
- **Disguised Digital Spaces:** Young women have developed sophisticated strategies for hiding SRHR and legal information within agricultural apps, prayer apps, and music files—turning the surveillance of shared devices into a site of resistance.
- **Community Repair Circles:** In several settlements, young women have formed repair collectives that fix devices using locally available materials, rejecting the dependency on male-dominated repair shops in town centers.
- **Elder-Youth Tech Partnerships:** Some communities are bridging the generational knowledge gap by pairing elder women with ecological knowledge and young women with digital skills, creating intergenerational tech teams that design pasture and water mapping tools.
- **Radio-as-Tech:** Community FM radio stations, powered by solar and managed by local women's councils, remain the most trusted information source. They are being integrated with digital content through offline broadcast-to-SD systems.
- **Data Refusal:** Increasingly, communities are refusing to participate in biometric data collection, demanding paper-based alternatives and organizing collective boycotts of surveillance-based aid distribution.

8. A FRAMEWORK FOR INDIGENOUS FEMINIST TECH SOVEREIGNTY

Based on our research and community consultations, Beyond Borders proposes the following framework for addressing tech challenges in the Northern Kenya corridor. This is not a call for more aid, more pilots, or more innovation competitions. It is a call for a fundamental shift in who controls technology, who benefits from it, and who decides what counts as innovation.

- **1. Infrastructure Justice, Not Connectivity Charity:** The Kenyan state and telecom companies must be held accountable for infrastructure apartheid. Universal service funds must be transparently directed to ASAL regions. Community-owned mesh networks and solar infrastructure must be recognized as legitimate public goods, not charity projects.
- **2. Data Sovereignty, Not Data Extraction:** All biometric and research data collection in indigenous communities must require free, prior, and informed consent with community-controlled governance. Indigenous communities must have the right to refuse data collection, demand deletion, and retain intellectual property over their knowledge.
- **3. Device Access as Bodily Autonomy:** Tech interventions must prioritize private, affordable, rugged devices for young women—not shared family phones. Device distribution must be accompanied by digital security training and safe charging infrastructure.
- **4. Indigenous Language Justice:** Tech platforms and tools must support Turkana, Samburu, Borana, Rendille, and other indigenous languages. This means keyboards, voice interfaces, auto-correct, and content creation tools in these languages—not just translation after the fact.
- **5. Community Archivist Model:** All tech projects in the ASAL must be led by community archivists—local young women trained in digital skills, content creation, and platform governance. External technical support must be temporary and transfer all control to community archivists within 18 months.
- **6. End the Innovation Industrial Complex:** Donors and funders must stop funding short-term tech pilots in the ASAL. Instead, they must support long-term, flexible, community-controlled infrastructure. Funding must include maintenance budgets, repair capacity, and exit strategies that leave communities stronger, not more dependent.
- **7. Recognize Indigenous Women's Tech Knowledge:** Elder women's ecological knowledge must be recognized as sophisticated data science. Tech design must start from this knowledge, not replace it. Indigenous women must be positioned as tech experts, not beneficiaries.

9. CONCLUSION: TECHNOLOGY AS TERRITORY

The tech challenges facing indigenous and pastoralist communities in Northern Kenya are not a digital divide waiting to be bridged. They are a form of ongoing colonial dispossession. The same forces that dispossessed pastoralists of their land—the state, the market, the aid industry—are now dispossessing them of their data, their knowledge, and their technological future.

For indigenous young women, this dispossession is doubly violent: they are excluded from technology by infrastructure apartheid, and they are harmed by technology through surveillance, extraction, and tech-facilitated GBV. Their bodies are data points for humanitarian algorithms. Their stories are raw material for Global North research. Their ecological knowledge is being digitized for carbon markets they will never benefit from.

But this report is not a story of victimhood. It is a story of resistance. Across the ASAL corridor, indigenous young women are building offline networks, disguising digital content, repairing their own devices, and refusing extractive data collection. They are not waiting for connectivity. They are creating autonomy.

The question for the feminist tech movement, for funders, for policymakers, and for technologists is simple: Will you support indigenous feminist tech sovereignty? Or will you continue to extract, pilot, and abandon? The future of technology in Northern Kenya will be determined not by the next innovation competition in Nairobi, but by whether indigenous young women are recognized as the architects of their own digital lives.

This report was created by the Beyond Borders collective through participatory research with 450+ community members across the Northern Kenya ASAL corridor. All findings were reviewed and validated by community archivists, women's councils, and elder advisors before publication. For more information, contact info@beyondborders.co.ke